NVLAP TESTING NVLAP LAB CODE:600177-0

GTG Global Testing Group

Test Report Of ANSI/IES LM-79-19

APPROVED METHOD FOR OPTICAL AND ELECTRICAL MEASUREMENTS OF SOLID-STATE LIGHTING PRODUCTS

Report Number..... : N02A23121042L00601

Client..... : RZ LIGHTING CO LTD

Address..... : 4F, Bldg. B, Zhongjing Sci-Tech Park, 606 Fengtang Blvd, Bao'an, Shenzhen, China 518103

Test Model..... : WP20-30L-[CS;CP;CSP]-3CCT

Brand Name..... : RZ

Testing Laboratory... : Guangdong Meide Testing Technology Co., Ltd.

Address..... : 1st floor, B Area, Jinbaisheng Industrial Park, Headquarters 2 Road,Songshan Lake Hi-tech Industrial Development Zone,Dongguan City,Guangdong Pr., China.

Testing Location..... : As above

Date of receipt..... : Mar. 16, 2024

Date of test ..... : Mar. 19, 2024 - Mar. 21, 2024

Date of report..... : Mar. 21, 2024

Tested by:	**Checked by:**	**Approved by:**
Allen chen	Jarvis zhang	
Allen Chen/ Test Engineer	Jarvis Zhang/ Project Engineer	Jessie Li/ Technical Manager

Guangdong Meide Testing Technology Co., Ltd. GTG

Note 1: The test data was only valid for the test sample(s).This test report is prepared for the customer shown above and for the device described herein. It may not be duplicated or use in part without prior written consent from Guangdong Meide Testing Technology Co., Ltd. This report must not be used by the customer to claim product certification, approval, or endorsement by NVLAP,NIST, or any agency of the Federal Government.
Note 2: This report does not imply product certification, approval, or endorsement by NVLAP, NIST,or any agency of the Federal Government.

1. Product Description for Equipment under Test(EUT)

Representative (Tested) Model:	WP20-30L-[CS;CP;CSP]-3CCT
Manufacturer:	RZ LIGHTING CO LTD
Product Type:	Outdoor Non-Cutoff and Semi-Cutoff Wall-Mounted Area Luminaires
Rated Voltage/Frequency:	120-277V AC, 50/60Hz
Rated Power:	15/20/30W
Rated luminous flux:	1950/2600/3900lm
Nominal CCT:	3000K/4000K/5000K
LED Manufacturer:	Lumileds Holding B.V.
LED Model No.:	L128-3080RC35004A1, L128-5080RC35004A1

2.Standards Used

- ANSI/IES LM-79-19:APPROVED METHOD:OPTICAL AND ELECTRICAL MEASUREMENTS OF SOLID-STATE LIGHTING PRODUCTS
- IES TM-30-18 IES Method for Evaluating Light Source Color Rendition (This Method is not in Nvlap accreditation scope)
- ANSI C82.77-10:2014 Harmonic Emission Limits – Related Power Quality Requirements for Lighting Equipment-Solid State

3.Test equipment list

Test Equipment	Serial No.	Model No.	Calibration due date
Full-field Speed Goniophotometer	MD-E028	GO-R5000	2024/09/16
Digital Power Meter	MD-E001	PF2010	2024/09/16
AC Testing Power Source	MD-E002	DPS1060	2024/09/16
Total Spectral Radiant Flux Standard Lamp	MD-E007	D908S	2024/09/25
Integrating Sphere System	MD-E029	2M	2024/09/16
High Accuracy Array Spectroradio Meter	MD-E011	HAAS-3000	2024/09/16
Digital Power Meter	MD-E008	PF310	2024/09/16
AC Testing Power Source	MD-E010	DPS1010	2024/09/16
Standard Lamp	MD-E036	D204	2024/09/25

Statement of Traceability: Guangdong Meide Testing Technology Co., Ltd. attested that all calibration has been performed using suitable standards traceable to national primary standards and International System of Unit(SI).

4. Test Method

Requirements of Ambient Condition
Product was tested with no seasoning. All stabilization and measurements were made in compliance with ANSI/IES LM-79-19. The product was operated at rated voltage or at voltage required by manufacturer. The ambient temperature of the sample was maintained at 25°C±1.2°C during measurement. And relative humidity between 10% and 65%.

Goniophotometer System
The sample was tested according to the ANSI/IES LM-79-19.

Photometric paramters were measured using a type C goniophotometer and software. The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, Luminous efficacy, zonal flux were calculated from the software taken at 1° vertical intervals and 22.5°horizontal intervals.Photometric distance was more than five times of the Largest dimension of the test SSL product.

Integrating Sphere System
The sample was tested according to the ANSI/IES LM-79-19.

The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

Fidelity Index (R_f) and Gamut Index (R_g) Calculation
The R_f, R_g was calculated according to IES TM-30-18 by using calculation tools. The calculation was based on the measured SPD from 380nm to 780nm with 1nm intervals. All the colors in this report is for reference only.

THD and PF Test

The sample was tested according to the ANSI C82.77-10:2014.

The sample was operated at rated voltage and was stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

5. Integrating Sphere Test Results

5.1 Test Data

Test Ambient Temperature (Integrating sphere internal temperature)	25.3℃	Test orientation	Downward
Operate time(Min.)	60	stabilization time(Min.)	30

Optical and Electrical Measurement Result

Mode	Voltage (V)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Luminous Flux(lm)	Efficacy (lm/W)	CCT (K)
30W-3000K	120.04	60	0.2474	29.11	0.9801	4018.8	138.06	3068
30W-4000K	120.05	60	0.2383	28.02	0.9795	4344.2	155.06	3960
30W-5000K	120.04	60	0.2459	28.93	0.98	4228.7	146.16	5112

Mode	Ra	R9	Rf	Rg	x	y	u'	v'	Duv
30W-3000K	82.6	6	85	96	0.4294	0.3963	0.249	0.5172	-2.05E-03
30W-4000K	85.3	18	85	95	0.3801	0.3701	0.2276	0.4985	-3.06E-03
30W-5000K	84	10	84	95	0.3421	0.3507	0.2098	0.4838	7.53E-04

5.2 Mode # 30W-3000K Color Rendering Index

Ra
82.6

R1	R2	R3	R4	R5
82	92	95	80	82

R6	R7	R8	R9	R10
91	81	58	6	83

R11	R12	R13	R14	R15
80	75	84	98	74

***5.3.1 Mode # 30W-3000K ANSI/IES TM-30-18 Color Rendition Report**

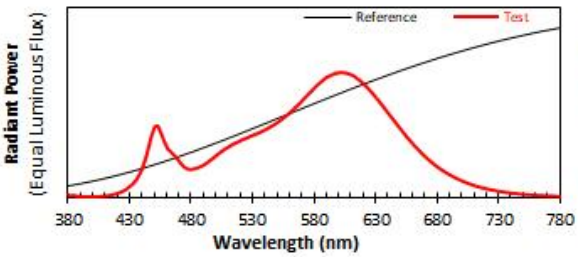
ANSI/IES TM-30-18 Color Rendition Report

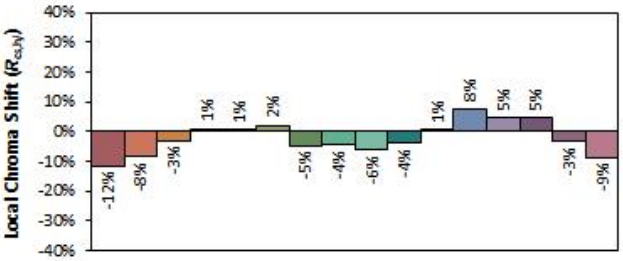
Source: L128-3080RC35004A1

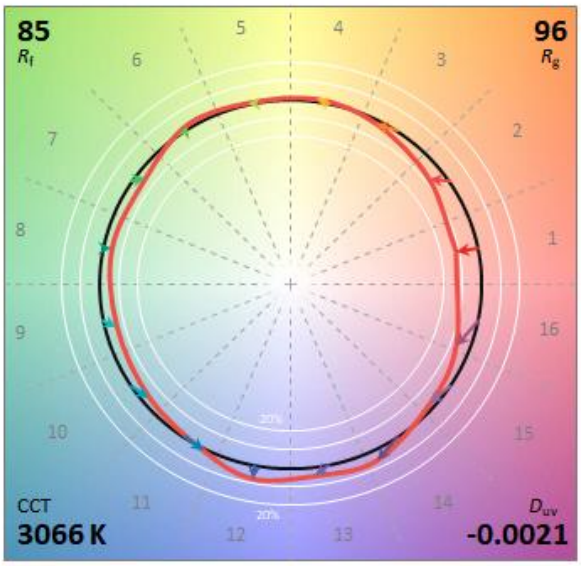
Manufacturer: RZ LIGHTING CO LTD

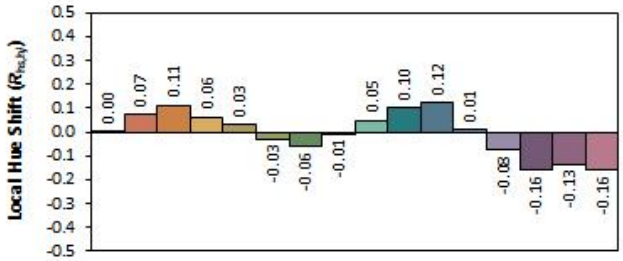
Date: 2024/3/19

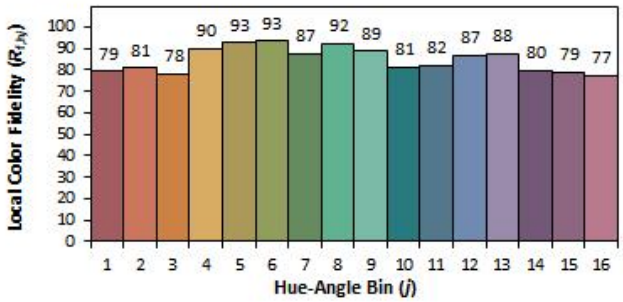
Model: WP20-30L-[CS;CP;CSP]-3CCT

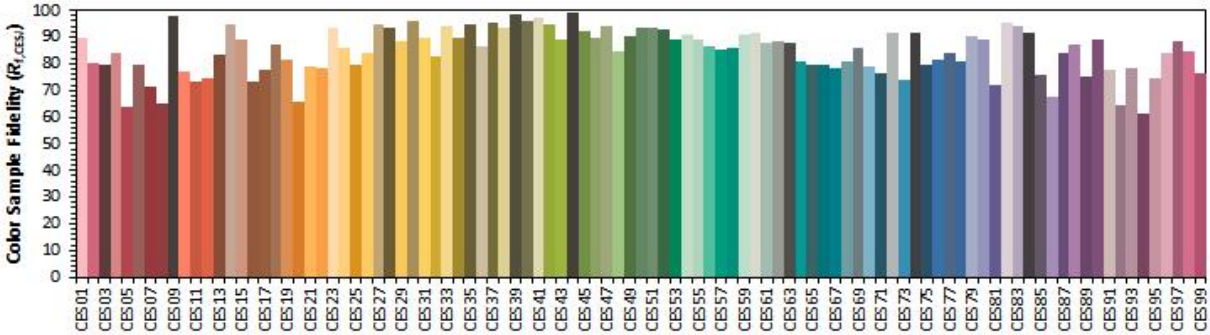












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	**0.4294**
y	**0.3962**
u'	**0.2491**
v'	**0.5171**

CIE 13.3-1995 (CRI)	
R_a	83
R_9	6

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

***5.3.2 Mode # 30W-4000K ANSI/IES TM-30-18 Color Rendition Report**

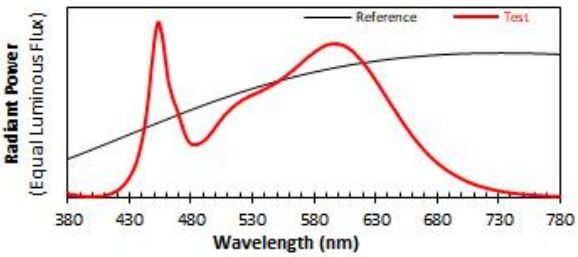
ANSI/IES TM-30-18 Color Rendition Report

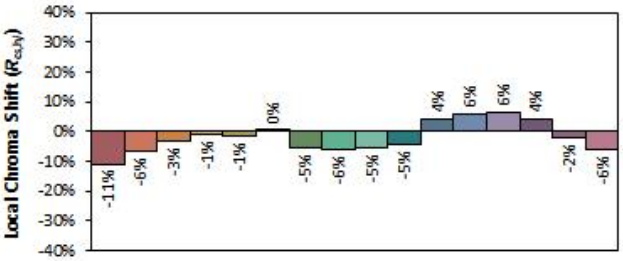
Source: L128-3080RC35004A1,L128-5080RC35004A1

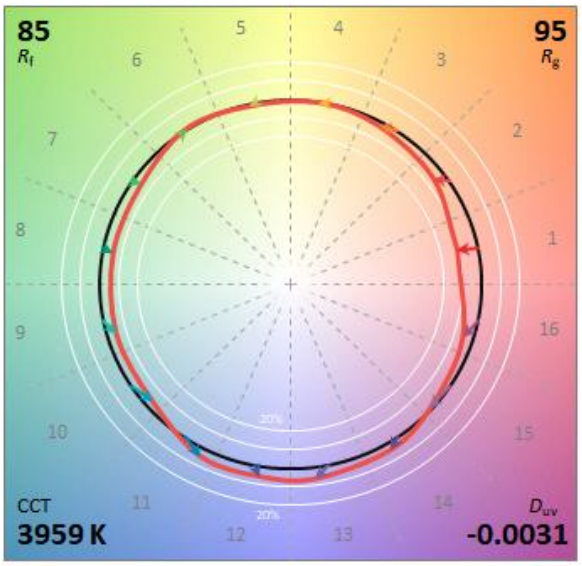
Manufacturer: RZ LIGHTING CO LTD

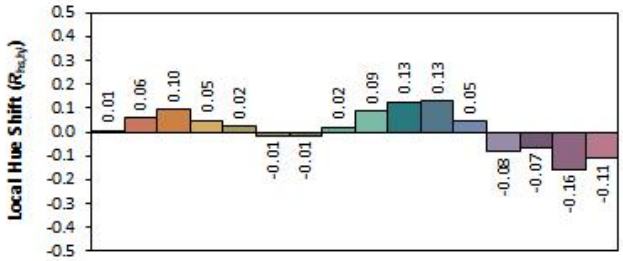
Date: 2024/3/19

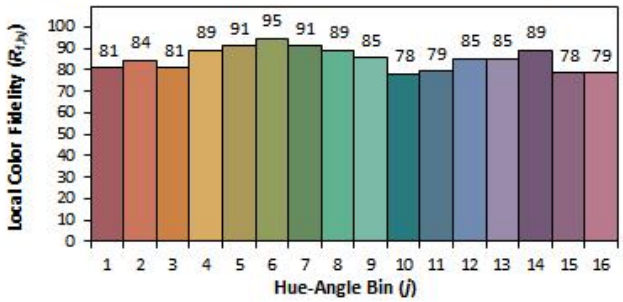
Model: WP20-30L-[CS;CP;CSP]-3CCT

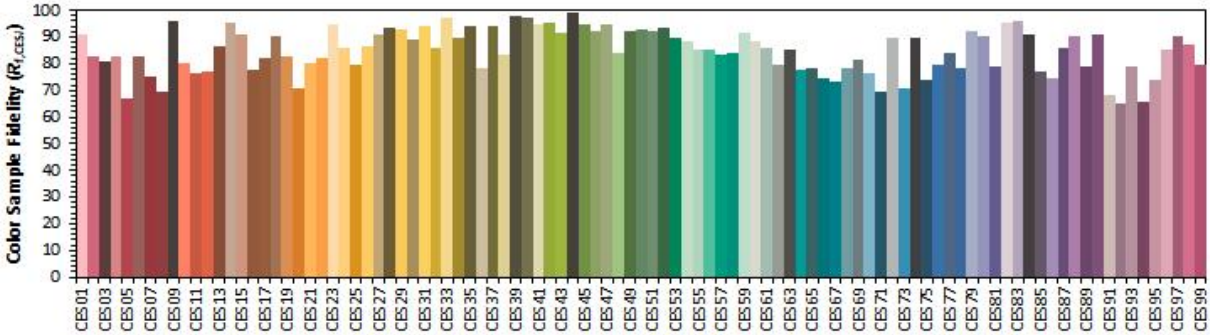












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	**0.3800**
y	**0.3699**
u'	**0.2276**
v'	**0.4985**

CIE 13.3-1995 (CRI)	
R_a	85
R_9	18

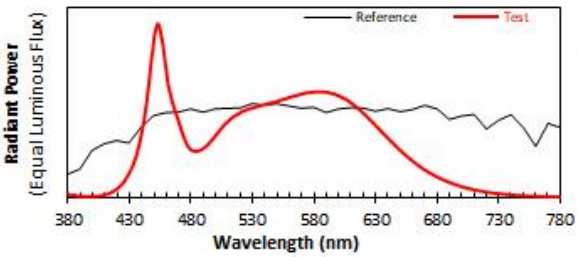
Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

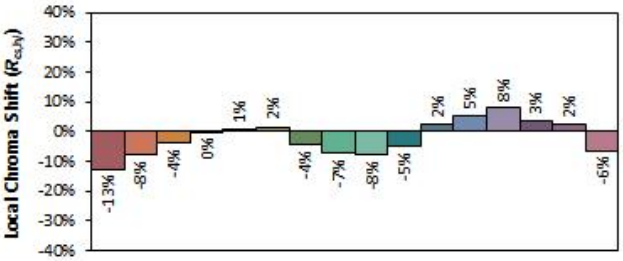
***5.3.3 Mode # 30W-5000K ANSI/IES TM-30-18 Color Rendition Report**

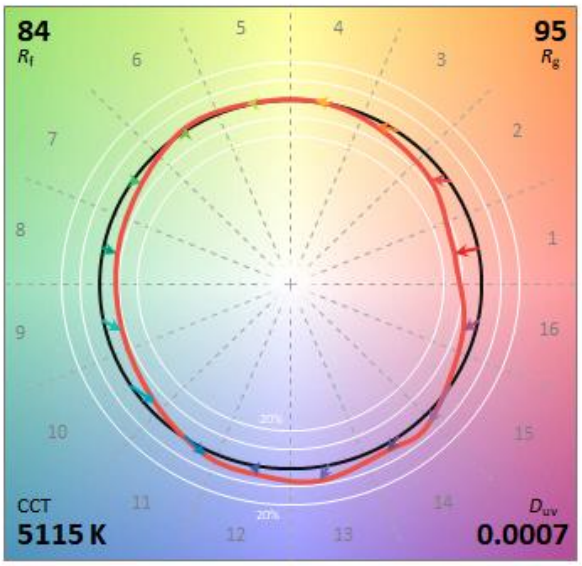
ANSI/IES TM-30-18 Color Rendition Report

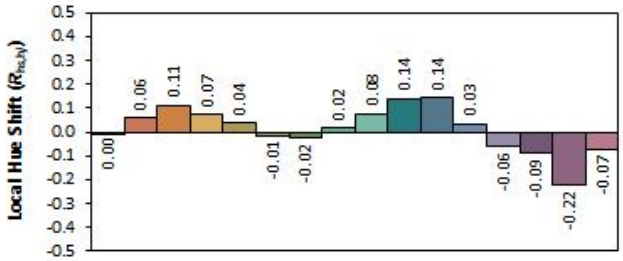
Source: L128-5080RC35004A1 **Manufacturer:** RZ LIGHTING CO LTD

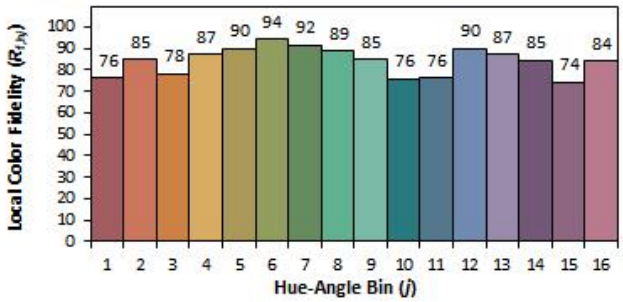
Date: 2024/3/19 **Model:** WP20-30L-[CS;CP;CSP]-3CCT

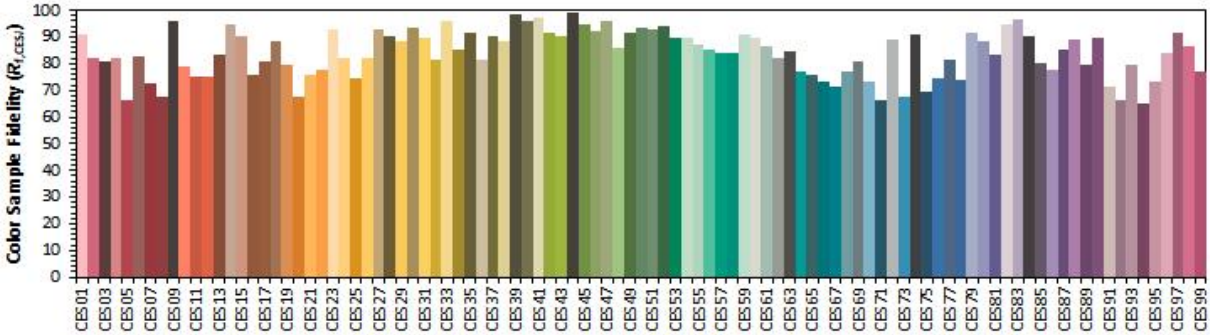












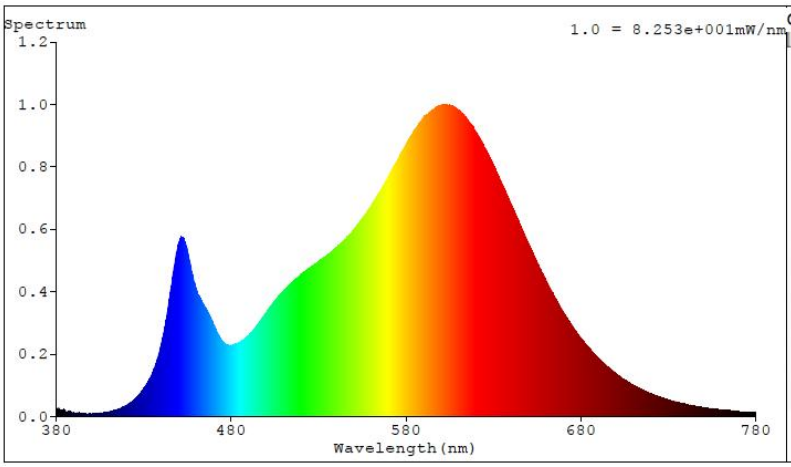
Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

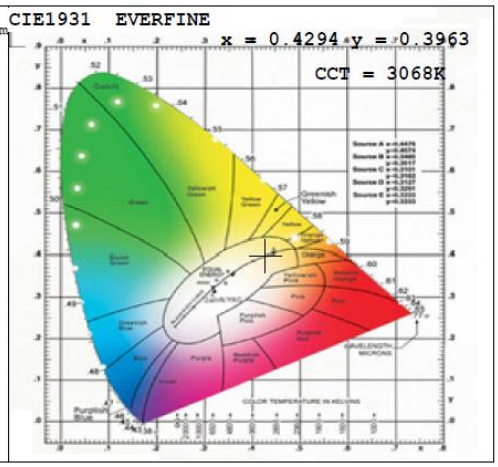
x	**0.3421**
y	**0.3505**
u'	**0.2098**
v'	**0.4837**

CIE 13.3-1995 (CRI)	
R_a	84
R_9	10

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

5.4 Mode # 30W-3000K Relative Spectral Power Distribution





nm	mW	nm	mW	nm	mW	nm	mW	nm	mW
380	0.0275	414	0.0204	448	0.5028	482	0.23	516	0.4363
381	0.0183	415	0.0225	449	0.5241	483	0.233	517	0.4402
382	0.016	416	0.0248	450	0.5566	484	0.2341	518	0.4455
383	0.0193	417	0.0271	451	0.5691	485	0.2382	519	0.4484
384	0.0175	418	0.0303	452	0.5736	486	0.2416	520	0.4543
385	0.0209	419	0.0321	453	0.5707	487	0.2459	521	0.4605
386	0.0135	420	0.0352	454	0.5544	488	0.2505	522	0.4648
387	0.0159	421	0.0397	455	0.5284	489	0.2542	523	0.4687
388	0.0095	422	0.0427	456	0.5041	490	0.2603	524	0.4713
389	0.0138	423	0.0468	457	0.4796	491	0.2662	525	0.4762
390	0.0099	424	0.051	458	0.4522	492	0.2711	526	0.4798
391	0.0099	425	0.0558	459	0.4313	493	0.28	527	0.4837
392	0.0095	426	0.0619	460	0.4112	494	0.2859	528	0.4908
393	0.0105	427	0.0667	461	0.3888	495	0.2951	529	0.4928
394	0.0074	428	0.0745	462	0.3816	496	0.302	530	0.4975
395	0.0073	429	0.0817	463	0.3707	497	0.3097	531	0.5013
396	0.0086	430	0.09	464	0.3626	498	0.3153	532	0.505
397	0.0094	431	0.0969	465	0.3512	499	0.3258	533	0.5088
398	0.0085	432	0.1062	466	0.3418	500	0.3319	534	0.5143
399	0.0092	433	0.1155	467	0.3322	501	0.342	535	0.5171
400	0.0079	434	0.1285	468	0.3201	502	0.3487	536	0.5222
401	0.009	435	0.1376	469	0.3087	503	0.3551	537	0.5266
402	0.0091	436	0.1531	470	0.2938	504	0.3655	538	0.5308
403	0.0104	437	0.1701	471	0.2824	505	0.3687	539	0.5361
404	0.0101	438	0.1873	472	0.2705	506	0.377	540	0.5404
405	0.0106	439	0.2074	473	0.2597	507	0.3868	541	0.5446
406	0.0102	440	0.2281	474	0.2494	508	0.3908	542	0.5508
407	0.011	441	0.2556	475	0.2425	509	0.3968	543	0.5562
408	0.0115	442	0.2829	476	0.2358	510	0.403	544	0.5631
409	0.0138	443	0.3142	477	0.2302	511	0.4079	545	0.5685
410	0.0149	444	0.3477	478	0.2289	512	0.4139	546	0.5717
411	0.0165	445	0.3874	479	0.2272	513	0.4213	547	0.5793
412	0.0172	446	0.4232	480	0.2284	514	0.4267	548	0.5861
413	0.0189	447	0.4637	481	0.2281	515	0.431	549	0.5901

nm	mW	nm	mW	nm	mW	nm	mW	nm	mW
550	0.5976	599	0.9915	648	0.5797	697	0.1512	746	0.0325
551	0.603	600	0.9972	649	0.5677	698	0.1471	747	0.0319
552	0.6131	601	0.9992	650	0.5543	699	0.1427	748	0.031
553	0.6204	602	0.9964	651	0.542	700	0.1375	749	0.0298
554	0.6267	603	0.999	652	0.5304	701	0.1335	750	0.0291
555	0.635	604	0.997	653	0.5167	702	0.1297	751	0.0282
556	0.6414	605	0.9954	654	0.5056	703	0.1256	752	0.0273
557	0.6486	606	0.9962	655	0.4928	704	0.1215	753	0.0265
558	0.6587	607	0.9912	656	0.4811	705	0.1175	754	0.026
559	0.6655	608	0.9892	657	0.4708	706	0.1138	755	0.0252
560	0.6752	609	0.9869	658	0.4587	707	0.1112	756	0.0244
561	0.6839	610	0.9808	659	0.4482	708	0.1068	757	0.0238
562	0.6929	611	0.9794	660	0.4364	709	0.1036	758	0.0231
563	0.7005	612	0.9726	661	0.4262	710	0.1004	759	0.0224
564	0.7129	613	0.9678	662	0.4141	711	0.0972	760	0.022
565	0.7202	614	0.96	663	0.4039	712	0.0941	761	0.0215
566	0.7291	615	0.952	664	0.3943	713	0.0912	762	0.0207
567	0.7392	616	0.9436	665	0.3824	714	0.0884	763	0.0199
568	0.7493	617	0.9381	666	0.3736	715	0.0854	764	0.0194
569	0.7582	618	0.9307	667	0.3632	716	0.0832	765	0.0189
570	0.7696	619	0.9212	668	0.3525	717	0.0808	766	0.0186
571	0.7802	620	0.9117	669	0.3427	718	0.0781	767	0.018
572	0.7914	621	0.9028	670	0.3338	719	0.0755	768	0.0174
573	0.8006	622	0.8943	671	0.3251	720	0.0735	769	0.0171
574	0.8107	623	0.8844	672	0.3161	721	0.0707	770	0.0165
575	0.8208	624	0.8739	673	0.3068	722	0.069	771	0.0159
576	0.8311	625	0.8614	674	0.2988	723	0.0669	772	0.0157
577	0.8405	626	0.8512	675	0.291	724	0.065	773	0.0153
578	0.8508	627	0.8418	676	0.2821	725	0.0625	774	0.0147
579	0.8607	628	0.8283	677	0.2743	726	0.0609	775	0.0147
580	0.8703	629	0.8184	678	0.2657	727	0.059	776	0.0141
581	0.8818	630	0.8048	679	0.2585	728	0.0572	777	0.0137
582	0.8924	631	0.7932	680	0.2515	729	0.0555	778	0.0132
583	0.8998	632	0.782	681	0.2439	730	0.0536	779	0.0134
584	0.9084	633	0.7708	682	0.2371	731	0.0521	780	0.0135
585	0.9184	634	0.7556	683	0.2302	732	0.0504		
586	0.9261	635	0.7442	684	0.2237	733	0.0489		
587	0.9345	636	0.7328	685	0.2168	734	0.0473		
588	0.9422	637	0.7194	686	0.2112	735	0.0458		
589	0.9494	638	0.7062	687	0.2042	736	0.0444		
590	0.9587	639	0.6952	688	0.1985	737	0.0431		
591	0.9638	640	0.6806	689	0.1931	738	0.0418		
592	0.9681	641	0.6697	690	0.1868	739	0.041		
593	0.9716	642	0.6551	691	0.1812	740	0.0393		
594	0.9782	643	0.6427	692	0.1759	741	0.0378		
595	0.9818	644	0.6297	693	0.1709	742	0.037		
596	0.9843	645	0.6173	694	0.1658	743	0.0359		
597	0.9915	646	0.6054	695	0.1606	744	0.0348		
598	0.9944	647	0.5917	696	0.1556	745	0.0337		

6. Goniophotometer Test results for mode # 30W-3000K

6.1 Test Data

Test Ambient Temperature	25.2℃	Test orientation	Downward
Operate time(Min.)	90	stabilization time(Min.)	30

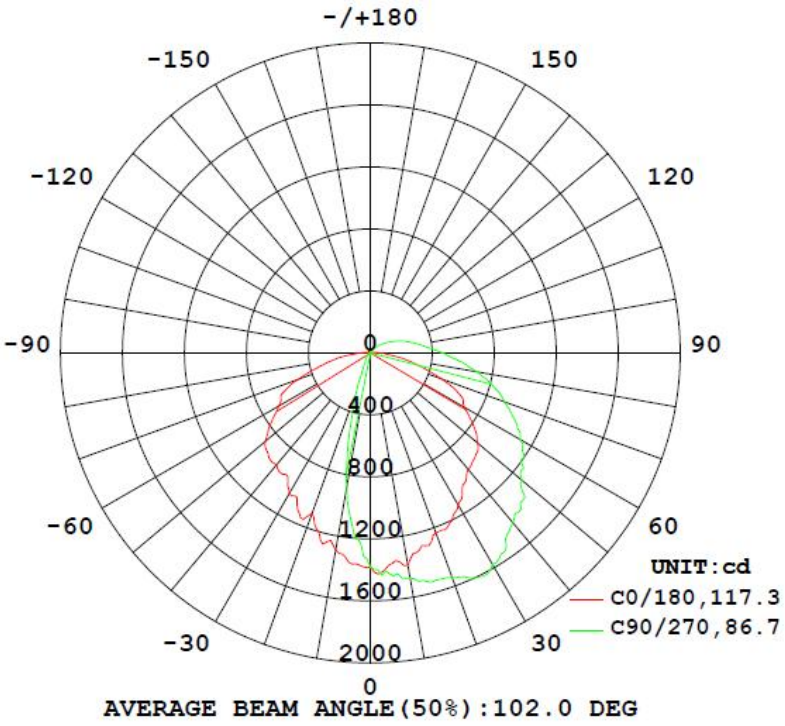
Electrical Measurement

Input Voltage (V)	Frequency (Hz)	Input Current(A)	Power Factor	Power(W)
120	60	0.2453	0.9881	29.08

Optical Measurement

Luminous Flux (lm)	Efficacy (lm/W)	Imax (cd)	Luminous Flux (0-90° zone)(lm)	Efficacy (0-90° zone)(lm/W)	ZL (80-90°)
4008.91	137.86	1613	3618	124.42	5.8%

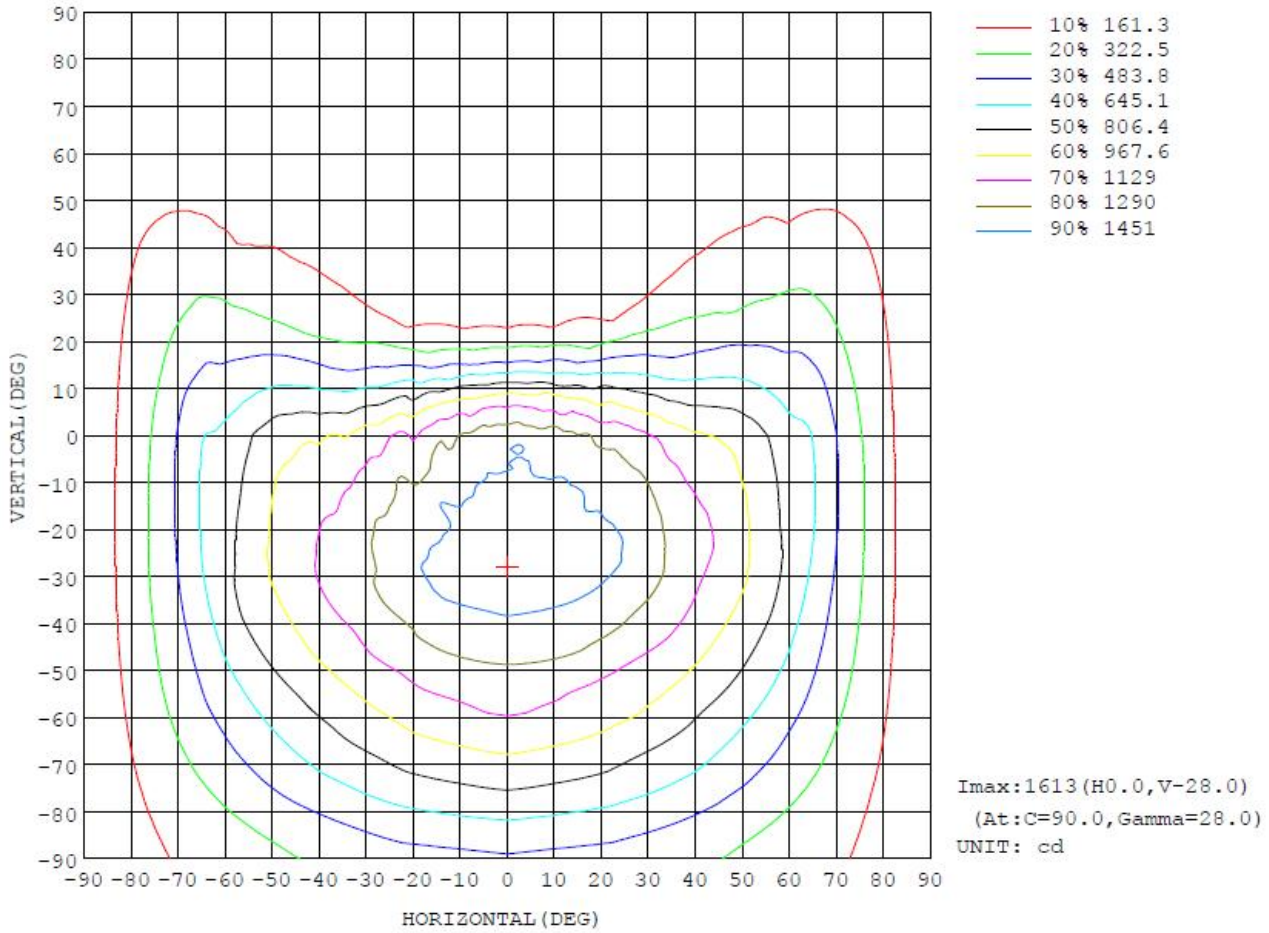
6.2 Luminous Intensity Distribution



6.3 Zonal Flux Diagram

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	Φ zone	Φ total	%lum,lamp
10	1390	1394	1475	1400	1292	998.8	880.0	1048	0- 10	124.2	124.2	3.1,3.1
20	1235	1487	1540	1464	1102	493.4	269.5	558.6	10- 20	315.9	440.1	11,11
30	1131	1480	1577	1395	1047	174.5	73.05	211.5	20- 30	435.8	875.9	21.8,21.8
40	982.4	1392	1432	1312	944.0	86.14	44.64	102.7	30- 40	519.7	1396	34.8,34.8
50	904.3	1169	1281	1156	885.3	64.56	23.88	76.52	40- 50	566.0	1962	48.9,48.9
60	699.5	968.1	1116	958.4	681.4	50.70	12.74	58.47	50- 60	555.2	2517	62.8,62.8
70	485.4	755.7	923.1	757.8	493.6	43.28	7.767	48.06	60- 70	493.2	3010	75.1,75.1
80	213.8	527.8	695.2	532.1	227.7	39.15	4.689	42.60	70- 80	372.4	3383	84.4,84.4
90	57.83	326.0	465.9	320.0	61.59	31.80	4.410	32.56	80- 90	235.3	3618	90.2,90.2
100	31.49	231.6	321.0	224.0	30.29	23.14	4.232	23.20	90-100	148.3	3766	93.9,93.9
110	14.10	161.7	224.7	158.4	17.81	16.55	3.490	16.39	100-110	100.8	3867	96.5,96.5
120	15.24	107.0	142.4	106.3	16.53	12.49	3.290	12.31	110-120	65.89	3933	98.1,98.1
130	12.27	67.74	81.22	68.91	13.77	9.438	3.289	9.242	120-130	39.15	3972	99.1,99.1
140	9.297	41.48	43.11	42.32	10.50	7.329	2.877	6.894	130-140	21.08	3993	99.6,99.6
150	6.512	22.09	23.32	22.73	7.566	6.237	2.658	4.672	140-150	10.37	4003	99.9,99.9
160	4.042	9.232	9.518	10.06	4.946	4.294	2.408	2.667	150-160	4.244	4008	100,100
170	2.155	2.127	1.534	2.503	2.729	2.388	1.829	1.707	160-170	1.132	4009	100,100
180	1.640	1.394	1.282	1.195	1.612	1.490	1.334	1.208	170-180	0.1583	4009	100,100
DEG	LUMINOUS INTENSITY:cd									UNIT:lm		

6.4 Isocandela Diagram



6.5 Luminous Distribution Intensity Data

Table--1 UNIT: cd

γ (DEG) \ C (DEG)	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5			
0	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382	1382			
5	1374	1410	1436	1445	1430	1405	1434	1390	1367	1291	1187	1210	1190	1232	1205	1307			
10	1390	1396	1394	1497	1475	1466	1400	1377	1292	1156	999	904	880	925	1048	1197			
15	1285	1380	1486	1530	1532	1483	1405	1378	1249	1021	771	569	515	611	802	1109			
20	1235	1385	1487	1517	1540	1491	1464	1265	1102	833	493	301	269	328	559	914			
25	1211	1372	1478	1554	1599	1520	1396	1317	1115	707	295	145	124	150	332	813			
30	1131	1362	1480	1564	1577	1536	1395	1245	1047	551	175	80.8	73.0	88.3	211	679			
35	1041	1284	1453	1494	1530	1485	1395	1202	956	439	117	58.6	55.7	62.3	139	545			
40	982	1225	1392	1406	1432	1384	1312	1150	944	360	86.1	48.0	44.6	51.4	103	456			
45	954	1178	1275	1336	1369	1309	1258	1129	932	322	72.2	37.6	32.5	41.4	86.6	399			
50	904	1097	1169	1224	1281	1246	1156	1104	885	274	64.6	28.2	23.9	31.9	76.5	345			
55	810	1012	1074	1126	1191	1143	1062	1014	781	231	59.0	21.3	17.6	24.6	67.3	288			
60	699	900	968	1050	1116	1054	958	909	681	204	50.7	15.7	12.7	18.8	58.5	245			
65	638	771	864	963	1031	969	849	793	635	166	46.1	12.1	9.74	14.8	52.8	200			
70	485	631	756	854	923	861	758	653	494	135	43.3	10.4	7.77	12.9	48.1	155			
75	332	499	640	757	819	760	645	514	341	104	41.5	8.71	6.02	11.2	46.0	111			
80	214	380	528	636	695	640	532	395	228	75.0	39.2	7.25	4.69	9.52	42.6	75.8			
85	110	266	412	525	569	527	408	271	121	51.2	35.3	6.37	4.10	8.43	37.3	48.9			
90	57.8	193	326	428	466	426	320	186	61.6	39.5	31.8	6.34	4.41	8.18	32.6	36.5			
95	37.9	159	274	353	379	352	266	151	41.7	32.3	27.4	6.14	4.48	7.70	27.5	29.5			
100	31.5	134	232	301	321	299	224	125	30.3	27.1	23.1	5.62	4.23	6.89	23.2	25.1			
105	12.7	115	194	253	270	251	189	107	17.3	24.2	19.5	5.10	3.85	6.11	19.4	22.2			
110	14.1	98.5	162	209	225	207	158	93.0	17.8	22.1	16.5	4.57	3.49	5.37	16.4	20.2			
115	15.1	83.6	133	169	182	167	130	79.5	17.7	20.4	14.2	4.22	3.31	4.85	14.1	18.8			
120	15.2	69.6	107	133	142	133	106	66.4	16.5	18.4	12.5	4.03	3.29	4.54	12.3	17.2			
125	13.9	56.7	85.4	102	109	102	85.8	54.3	15.1	16.7	10.8	3.95	3.40	4.26	10.7	15.8			
130	12.3	45.3	67.7	77.3	81.2	77.0	68.9	44.1	13.8	15.1	9.44	3.66	3.29	3.86	9.24	14.4			
135	10.7	35.8	53.6	57.4	58.9	57.7	54.6	35.3	12.0	13.4	8.26	3.38	3.08	3.42	8.00	12.8			
140	9.30	27.9	41.5	43.0	43.1	43.7	42.3	27.9	10.5	11.5	7.33	3.13	2.88	2.99	6.89	10.9			
145	7.90	21.2	31.0	32.5	32.0	33.1	31.7	21.6	8.89	9.42	6.75	3.02	2.73	2.66	5.80	8.98			
150	6.51	15.4	22.1	23.8	23.3	24.4	22.7	16.0	7.57	7.67	6.24	3.11	2.66	2.50	4.67	7.20			
155	5.20	10.9	14.8	16.2	15.9	16.9	15.5	11.8	6.23	6.45	5.38	3.20	2.59	2.45	3.54	5.52			
160	4.04	6.67	9.23	9.96	9.52	10.6	10.1	7.95	4.95	5.11	4.29	3.01	2.41	2.32	2.67	3.96			
165	3.04	3.73	4.79	4.78	4.41	5.14	5.50	4.43	3.79	3.81	3.30	2.46	2.08	2.01	2.03	2.68			
170	2.15	2.17	2.13	1.85	1.53	1.80	2.50	2.36	2.73	2.73	2.39	1.88	1.83	1.79	1.71	1.89			
175	1.62	1.51	1.35	1.23	1.18	1.17	1.15	1.27	1.83	1.84	1.70	1.59	1.56	1.51	1.47	1.46			
180	1.64	1.51	1.39	1.34	1.28	1.21	1.20	1.21	1.61	1.62	1.49	1.39	1.33	1.29	1.21	1.20			

7. THD and PF Test

Mode	Voltage (V AC)	Frequency (Hz)	Power Factor	THD (%)
30W-3000K	120.0	60	0.980	11.53
	277.0	60	0.895	14.56
30W-4000K	277.0	60	0.889	15.06
30W-5000K	277.0	60	0.894	14.53

8. Photo of sample

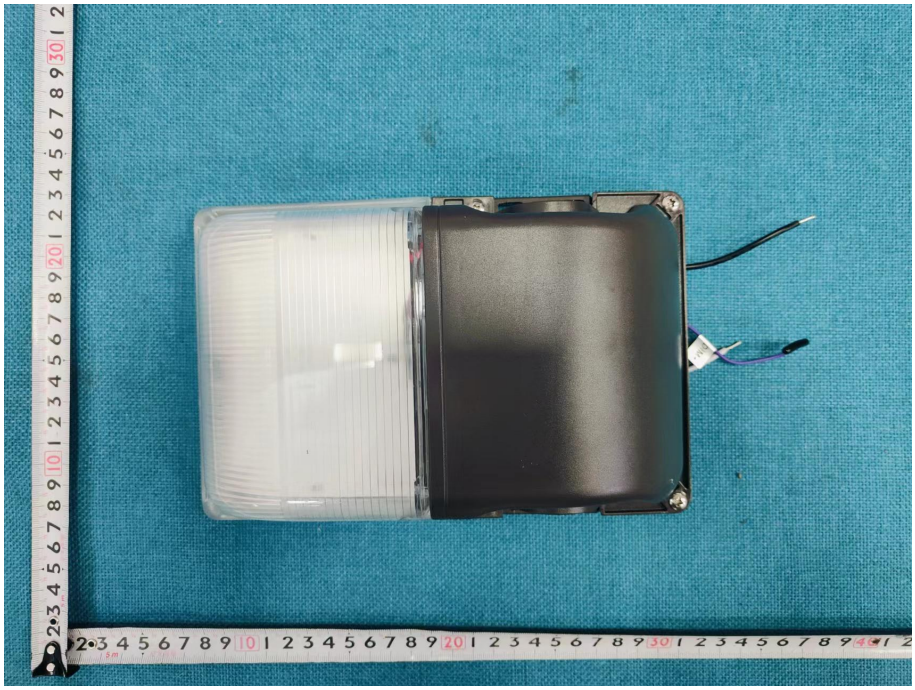


Figure 1

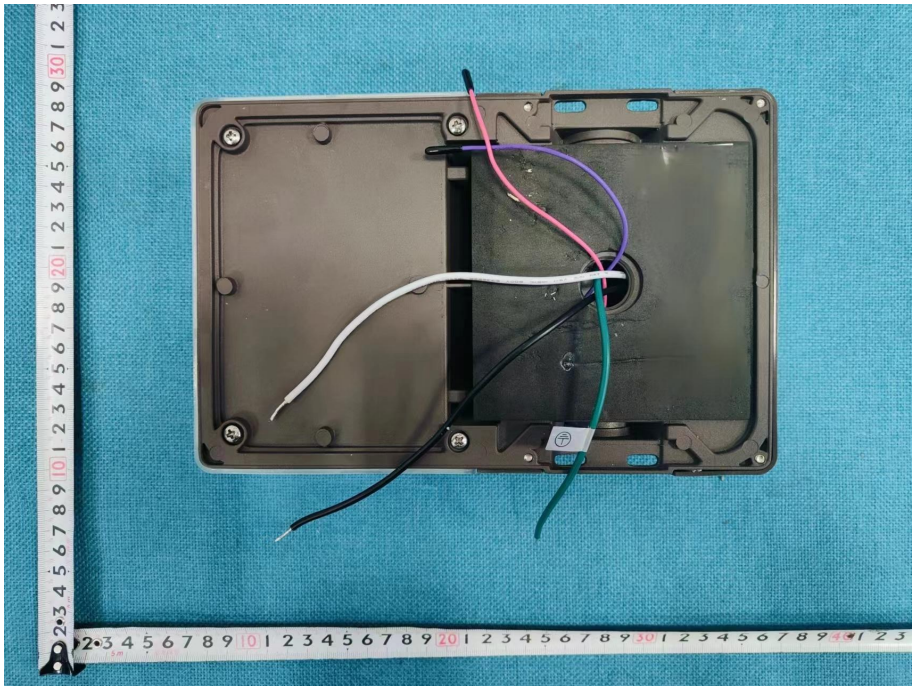


Figure 2

---End of Report---